

Computer and Network Security

PROJECTS

Description

The project of computer and network security course is a precious opportunity to convert the acquired theoretical knowledge into practical solutions that enhance the security of information systems and face off the various existing security threats. Also, it will be an occasion to teamwork and cooperate with your peers to build solid solutions. This will allow developing your teamwork skills, as per management, task distribution, communication, and cooperation.

To Do

In this project, you must choose two topics: One from the primary list and another one from the secondary list (see Pages 2 to 5).

Requirements

You are required to do the following five tasks:

1. Finalize a primary and a secondary project.
2. Prepare some Latex slides to present the two projects,
3. Prepare a project report — choose whatever template you like from Overleaf — discussion your project problematic, your solution, implementation, and the obtained results,
4. Prepare a demonstration in the form of a recorded video and upload it over Youtube. The video demo isn't just a technical showcase; it's a communication exercise. Don't use AI voiceover and dubbing.
5. Project submission: zip your project resources (report, slides, source code, link to videos, etc), upload the file to Google Drive, and share the link with the teaching team.

Firm Deadline to Submit Chosen Projects

The deadline is fixed for Friday, **Friday— Feb 20th, 2026 — 11:59PM**.

Firm Deadline to Submit your Final Work

The deadline is fixed for Friday, **Friday— May 08th, 2026 — 11:59PM**.

Presentation and Demos

You will present and demonstrate your work on Week 14 and 15 during lab sessions.

Primary Projects

- **[P000] Secure student face recognition system.** This system adopts deep learning technology to develop an intelligence face recognition system for students. This, for example, can be used to track student attendance, or perform authentication. The system should provide a level of security where a malicious person cannot use a picture of the student or a crafted mask to bypass the system.
- **[P001] Next-generation firewall (NGFW).** In this project, we focus on content detection and connection filtering. The system will use machine learning or deep learning technology to perform deep packet inspection, detect content without decryption (e.g., VPN content), and filter related connections.
- **[P002] Voice Recognition-Based Multi-Factor Authentication System.** Implement a voice recognition-based Multi-Factor Authentication system using an Arduino/RaspberryPI and a suitable microphone sensor. The system will require a user to repeat a unique (unpredictable), randomly generated sentence to authenticate and gain access (e.g., open a door 672). This prevents attackers from using a recorded voice to bypass the system. The system relies on classical machine learning or deep learning technologies to train an artificial intelligence.
- **[P003] Behavioral Biometric Authentication System.** This project develops an AI-based authentication system that identifies users through their behavioral patterns such as typing rhythm, mouse movements, and touchscreen gestures. The system learns each user's normal behavior and continuously monitors activity to detect impostors. Even if a password is stolen, attackers cannot easily imitate behavioral traits. The work involves machine learning-based user profiling, anomaly detection, and evaluation using metrics like FAR and FRR. It supports continuous, more secure authentication for sensitive systems.
- **[P004] Malware.** Here you will develop malware that can compromise various security services. The malware should evade most anti-malware tools.
- **[P005] Homemade RFID proxmark for MIFARE 13.55MHz.** This project consists of modifying the MFRC522 RFID reader (Arduino module) to be able to read RFID MIFARE cards from a longer range (1.5 to 2m, or more). The project requires a fundamental understanding of electronics and antenna design. You will be given an MFRC522 RFID reader to do the project.
- **[P006] Side-Channel Attack on Arduino Cryptography.** In this project you will demonstrate how cryptographic implementations can leak secret information through physical side channels, even when the encryption algorithm itself (e.g., AES) is mathematically secure. You will implement the AES encryption algorithm on an Arduino board and perform a power analysis attack to recover the secret encryption key. Then, design and evaluate countermeasures to mitigate the attack.

- **[P007] Privacy-Preserving Face Recognition using Federated Learning.** This project develops a face recognition system where student biometric data never leaves their personal device. Instead of collecting all images on a central server, the system uses federated learning to train a shared model while keeping raw images locally stored. The objective is to reduce privacy risks related to biometric databases. Students will implement local model training, secure aggregation, and evaluate resistance against data leakage attacks (e.g., model inversion). The project includes performance–privacy tradeoff analysis.
- **[P008] Differential Privacy-Based Student Analytics Platform.** This project designs a student performance analytics dashboard that integrates differential privacy mechanisms. The system adds calibrated noise to statistical outputs (e.g., average grades, attendance rates) to prevent re-identification of individuals. Students will implement Laplace or Gaussian mechanisms, measure privacy budget (ϵ), and evaluate utility degradation. The final system demonstrates how institutions can publish useful statistics without compromising individual privacy.
- **[P009] AI Model Inversion Attack and Defense.** This project demonstrates how trained machine learning models can leak private training data through model inversion attacks. Students will train a simple classifier (e.g., facial or medical dataset), attempt to reconstruct sensitive inputs from the model, and then implement defenses such as differential privacy or regularization. The goal is to understand privacy risks in AI systems.
- **[P010] Cover Channel through DNS Tunneling.** This project explores the exploitation of the Domain Name System (DNS) as a covert communication channel to bypass restrictive firewall policies and facilitate unauthorized data exfiltration. By encapsulating non-DNS traffic within standard queries and responses, attackers can leak sensitive information under the guise of legitimate name resolution—a technique known as DNS tunneling. To counter this stealthy threat, the project implements a Machine Learning (ML)-based detection framework that analyzes DNS traffic patterns in real-time. By extracting key statistical features such as subdomain entropy, query frequency, and payload length, the system employs anomaly detection algorithms to distinguish between benign requests and malicious tunnels, providing a robust mechanism for the detection and mitigation of data breaches that traditional signature-based security tools often overlook.

Secondary Projects

- **[S001] Cybersecurity Awareness Program.** Develop a web application (e.g., or a web-app) where employees of the school and students can use it to solve security challenges, to train themselves and become more aware and sensitive. The users go through periodic quizzes where they are challenged to analyze some security-related problems and answer some questions. The score is sent to their email account. At the same time, the tool shows some statistics about the security awareness of the school. Your application can be used in any company to develop its social security aspect.
- **[S002] Tiny Intrusion Detection System (tIDS).** In this project, you will use classical machine learning algorithms to implement a LAN-based intrusion detection system focusing on the detection of the following seven attacks: (1) STP spoofing attacks, (2) ARP poisoning, (3) DNS spoofing, (4) TCP SYN flood, (5) Smurf attack, (6) IGMP spoofing, and (7) UDP flooding.
- **[S003] Simple Web Vulnerability Scanner.** This project aims to develop a basic web security assessment tool designed to identify common vulnerabilities in web applications, including SQL injection, Cross-Site Scripting (XSS), and the presence of outdated or insecure components. The scanner automatically analyzes web pages, detects potential security weaknesses, and produces a structured report to assist administrators in understanding and mitigating risks. The work requires a solid understanding of web application architecture, HTTP communication, and typical web attack vectors. You will design vulnerability detection modules, implement scanning mechanisms using Python or JavaScript, and evaluate the effectiveness of the tool within controlled test environments.
- **[S004] Ransomware Detection and Mitigation System.** This project aims to design an intelligent security system that detects and mitigates ransomware attacks using machine learning techniques. The system monitors system and network behavior in real time to identify anomalies associated with encryption activities and malicious processes. It integrates detection models with rapid response mechanisms such as process isolation and file protection to reduce damage. The work involves feature extraction, anomaly detection, and model evaluation on simulated attack scenarios.
- **[S005] Phishing Email Detector using Machine Learning.** This project develops a system that detects phishing emails using machine learning and text-based features. You will have to extract indicators such as suspicious URLs, keywords, and email structure, then train and evaluate classification models. A small interface will be designed to demonstrate phishing probability prediction.
- **[S006] Email Spoofing Detector.** In this project you develop an email spoofing browser extension. The tool will access the email header and analyze the different information, including, SPF, DKIM, DMARK, timezone, content, etc, to help the user detect possible phishing emails.

- **[S007] Software-based Jammer.** In this project, you are going to develop malware. This malware consists of a Wi-Fi radio jamming tool using just software and a Wi-Fi network card. The purpose is to show how using a simple packet-injection capable network card, you can permanently occupy the radio by injecting random Wi-Fi frames and fooling others to believe the radio is being used by legitimate users for a short period of time.
- **[S008] Fingerprint Bypassing System.** In this project, you are asked to first build an authentication system that relies on fingerprint verification. You will be provided with an Arduino microcontroller and the fingerprint scanner module FPM10A. Then, you are asked to craft/make customized fingerprints (e.g., silicon-based fingers, tap-based fingerprints, etc) to bypass the authentication system. If your findings prove to be successful, you may be invited to try your attack on the school's fingerprint authentication system.
- **[S009] Secure and Private E-Voting Prototype.** This project designs a small-scale electronic voting system that guarantees vote confidentiality, integrity, and anonymity. Students implement cryptographic primitives such as homomorphic encryption or blind signatures to ensure that votes are counted correctly without revealing voter identity. The system includes a threat model and attack evaluation.
- **[S010] Smart Home Privacy Monitor (IoT Traffic Analysis).** This project analyzes encrypted IoT traffic from smart home devices to determine whether user activities (e.g., sleeping, leaving home) can be inferred from traffic patterns. Students then implement countermeasures such as traffic shaping or packet padding. The project highlights privacy risks even when communication is encrypted.